

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MRBENCH | Context: Indian Metropolitan Grade 1–8 Mathematics Teachers (Delhi & Mumbai)

Definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Justification

The benchmark is text-only English [Q133, Q134, Q147], matching the deployment's text-based English mobile/enterprise application interface. The target population is fluent in English professionally. No script mismatch, no multimodal mismatch, and adequate infrastructure (urban Tier-1 internet penetration ~80–88%, smartphone access near-universal for professional teachers) [WEB-5, WEB-6, WEB-7] support text-based English signal delivery. Minor concern: no multimodal capture of handwritten student work, which is common in actual Indian classrooms, but the evaluator-role deployment (teachers judging text dialogues) does not require this. The user weighted IF as LOWER priority.

ID	Evidence
WEB-5	Urban Tier-1 internet penetration ~88%, broadband >50% in urban households
WEB-6	India had 660 million smartphone users as of 2024–2025
WEB-7	National mobile data median speed 94.62 Mbps in early 2024

Strengths

- Text-only English [Q133, Q134] matches deployment interface exactly
- No script or modality mismatch — both benchmark and deployment are Latin-script English
- Target population has fluent English and reliable mobile infrastructure [WEB-5, WEB-7]

ID	Evidence
WEB-5	Urban Tier-1 internet penetration ~88%, broadband >50% in urban households
WEB-7	National mobile data median speed 94.62 Mbps in early 2024

- Length and turn metadata documented per dataset [Q141, Q142, Q147]

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distributions match: text-based English dialogues [Q133, Q134] correspond to deployment's text-based English interface. No image, audio, or specialized capture parameters required.

IF-2: Regional infrastructure supports the format: urban Indian metro internet penetration is high (~80–88%) and smartphone access is near-universal for professional teachers [WEB-5, WEB-6].

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IF-3: Domain-specific form differences are minimal. The benchmark does not include images of handwritten work [no quote], which is common in Indian classrooms; however, the evaluator-role deployment (teachers judging text-based AI+human responses) does not require multimodal input.

IF-4: No form mismatches significantly harm external validity for this deployment. Conversation length differences between Bridge and MathDial [Q141, Q142] are documented and do not introduce mismatch with the deployment.